

- The internet and Network Fundamentals**
- Network Edge and Network Core**
- Communication and Network Protocols**
- Understanding Network Performance**
- Network Security**

- What a computer network is.
- Collection of devices, routers, and links, all of which is managed by an organization of sorts, for the sake of running a computer.
- Why the Internet is commonly described as a "network of networks."
- Made up of Interconnected ISPs, each of which is a network
- (internet service providers)
- How billions of devices communicate through interconnected networks.
- Protocols EVERYWHERE, for getting things to and from various computers
- Examples of end systems or hosts:
- Make up the network's edge
- laptops, smartphones, servers, IoT devices, and cloud systems.
- Give at least two real-world examples of how you personally interact with computer networks
- Laptop, Smart Watch

- ~~Explain the difference between the network edge and the network core.~~
- Edge is how people get TO the internet
- Core is The internet itself, Wifi, Routers, the "Network of Networks"

- Hosts
- End systems
- Clients
- A type of host (desktops, laptops, smartphones, etc.)
- Servers
- Machines that store and distribute web pages, stream videos or games, relay emails, etc.
- Access networks
- The network that physically connects the end system to the router
- Home networks
- Household internet/appliances
- Enterprise networks
- Business's computers
- Mobile networks
- Mobile phones

- Routers
 - Network core
 - Switches
 - Forward packets to the intended destination
 - Links
 - Links transmitter and receiver, access networks
 - Packet forwarding
 - The process of determining which link the router will forward the packet to
 - Packet switching
 - A method of sending data to and from different computers by splitting them into smaller packets
 - (restaurant with no reservations, but you have to wait for your table maybe)
 - Circuit switching
 - Similar to packet switching, though the needed resources are now reserved for the travelling data
 - (restaurant with only reservations that need to be set up beforehand)
 - Explain how a packet can travel from one device to another through multiple network devices.
 - Routers, Switches, and Links essentially work together to create a road that tells the packets of data where to deliver its cargo
- Client -> Access Network -> Router -> Internet/Core -> Router -> Server
- Explain what a network protocol is and why protocols are necessary.
 - A conversation between two communicating remote entities that dictate how they interact.
 -
 - HTTP/HTTPS (HyperText Transfer Protocol (Secure))
 - Web Document request and transfer
 - Application Layer
 - Sending Emails between clients and mail servers
 - TCP (Transmission Control Protocol)
 - Connection-oriented transport protocol. Has sender/reciever speed messaging, breaks long messages into shorter ones, and congestion control (throttling transmission rates when the network is congested)
 - Transport Layer
 - Google uses it to Send information between its private data centers
 - IP (Internet Protocol)
 - Specifies the format of packets sent and received among routers and end systems
 - Network Layer

- Google uses it to Send information between its private data centers
- DNS (Domain Name System)
- Translates a given hostname into it's IP address
- Application Layer
- Taking a student to Youtube after entering it's URL by turning the host name into youtube's IP address.

- You want to go to Youtube.Com
- A student starts by entering it's host name, or URL
- DNS to grab the needed IP address
- IP address to prepare the road to it
- TCP prepares for transport
- HTTP sends the official request to transfer over the needed data

- Throughput
- The amount of data per second that can be transferred
- Often constrained, because not every system can handle instantaneous transmission
- Bandwidth
- The width of the frequency band
- The higher it is, the more things it can connect
- Delay
- The temporary halting of a packet's journey at various nodes.
- These can accumulate, causing a total nodal delay
- Transmission delay
- A delay caused by a first come, first serve system, where it has to wait it's turn in order to transmit itself
- Queuing delay
- Delays to the arrival of the packets on account of network congestion
- Packet loss
- A packet is dropped due to a full buffer

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What command or tool you used.

What information it produced.

What the result means.

How the result relates to something discussed in class.

comparing network performance under different conditions

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What command or tool you used.

What information it produced.

What the result means.

How the result relates to something discussed in class.

comparing network performance under different conditions

- Explain why security is essential in modern computer networks.
- Pretty much everything about a person can be found online, so it's more important than ever that this info stays secure

- Your discussion must include:

- Authentication
- Making sure only specific sources can participate in a certain protocol
- Encryption
- Scrambling a message in various ways, so that only those with the key can read it
- Confidentiality
- Preventing information from being seen by those without permission to do so
- Network threats
 - ...threats to the network
- IP spoofing
- Injecting packets with false source addresses
- Packet sniffing
- A passive receiver that records a copy of any packets it can spot

- Use at least one real-world security scenario to illustrate your explanation.

- Firewalls
- Filter select incoming packets
- VPNs
- Encrypts data sent over the internet
- Secure Authentication
- Log in before you can even use anything, plus multi-factor authentication for the software itself

- Thank you for listening, and have a great rest of your day.